Macro News Pipeline

Direct-link relevance framework

April 16, 2026

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Outline

Motivation: DJNW metadata is unreliable

DJNW tagging scheme: full census

The direct-link relevance rules

Generator-verifier architecture

Results on DJNW March 2022

Example mappings: before / after

Open design questions

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Rules for a valid asset mapping

An article–asset pair is relevant **only** when the asset is affected through one of five direct mechanisms. These rules are intentionally *asset-class-agnostic* — the same five cases apply to commodities, currencies, rates, equity indices, and sectors. In the future we may layer asset-class-specific rules on top (e.g., different thresholds for what counts as a “dominant input” for base metals vs. energy), but the current framework is general.

1. **NAMED** — the asset appears in the text as a subject.
2. **PRIMARY VEHICLE** — the asset is the dominant traded instrument for a specific force described in the article.
3. **ISSUING AUTHORITY** — the article’s subject is the institution that controls this asset’s price.
4. **SUBSTITUTION / INPUT LINK** — a named commodity and this asset are closely-related substitutes or inputs.
5. **SECTOR CONSTITUENT** — a named company operates in this equity *sector* (not index).

The following slides walk through each rule with a concrete DJNW example.

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Case 1: NAMED

Rule. The asset is named in the text as a *subject* affected by a specific force — not merely as a pricing unit, reporting currency, denominating proxy, or incidental mention.

Example: "ICCO Cocoa Daily Price — Mar 21"

The article reports the daily closing price of cocoa. Cocoa futures → **strong**, relevance 1.0.

The article originally also mentioned prices *"in Euro/ton."* Euro is a pricing unit here, not a subject — rejected under case 1's exclusion clause. This led us to add an explicit *functional-currency / reporting-currency* rejection rule after the same pattern leaked on the Carnival earnings article.

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Case 2: PRIMARY VEHICLE

Rule. The asset is the dominant traded instrument for the economic exposure described in the article. The exposure must be a *specific force* — not “growth,” “uncertainty,” or “risk-off.” The asset may be listed in a different jurisdiction than the named force.

Example: “China Jan-Feb Property Investment Rose 3.7%”

The article describes a slowdown in Chinese property construction. The **Hang Seng Index** is the primary listed market for taking direct equity exposure to the Chinese economy — accepted as **weak 0.60**.

Without the cross-jurisdictional clause, the validator had rejected HSI by inventing “the Hong Kong economy” as a fake intermediate step. The safeguard we added stops the validator from fabricating intermediates.

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Case 3: ISSUING AUTHORITY

Rule. The article's subject is the central bank, government agency, or supranational body that issues, controls, or directly sets this asset's price.

Example: "BOE's Rate Decision Could Benefit Sterling"

The Bank of England is the issuing authority for UK monetary policy. **UK Long Gilt** → weak 0.60, **3-Month SONIA** → weak 0.85. Sterling itself comes in via case 1 (named, strong 1.00).

Case 3 also drives the full UST curve on the *"Dollar Firms, Ukraine War"* article, where the Fed's tightening stance is the named subject and the Fed is the issuing authority for US government debt.

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Case 4: SUBSTITUTION / INPUT LINK

Rule. The article names a commodity, input, or product, and this asset trades a closely-related substitute, input, or complement. The relationship need not be stated in the article — well-known industry relationships count, as long as the original commodity is named.

Example: “China Jan-Feb Property Investment” (cont.)

The article names *“new construction starts.”* **LME Copper** (weak 0.50) and **LME Zinc** (weak 0.45) are accepted because Chinese construction is the dominant global demand driver for both metals.

LME Aluminium and LME Lead are *rejected* on the same article — property construction is not a dominant demand source for either. The guardrail is the word *dominant*: “any industrial use” is not enough.

Signal: always **weak**. **Future work:** asset-class-specific rules may refine the “dominant” threshold for base metals vs. energy vs. agriculture.

Case 5: SECTOR CONSTITUENT

Rule. The article names a company or business event, and this asset is an equity **sector** (not an equity index) where the named company operates. Case 5 is restricted to sectors whose composition is defined by industry classification.

Example: "Carnival 1Q Loss/Shr $1.66"

Carnival Corporation is a major cruise-line operator. **Consumer Discretionary Sector** → weak 0.65.

Equity *indices* (S&P 500, Nasdaq 100, DAX, etc.) are excluded from case 5 because LLMs cannot reliably recall index composition. On the Russia/Boeing article, the model hallucinated that Boeing is a Nasdaq 100 constituent — it is not. Indices can still be tagged via cases 1–4 (e.g., HSI on China via case 2).

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What does not count as a direct link

Each exclusion was motivated by a specific pattern the pipeline proposed that was judged incorrect upon review.

Exclusion rule	**Motivating example**
Generic “risk-off / safe-haven / global growth” chains	Gold, JPY, Swiss franc tagged on every geopolitical article
Historical correlations without article-specific pathway	“Commodities tend to move with the dollar” as standalone justification
Secondary derivations through unnamed intermediates	China property $\rightarrow$ “Australian economy” $\rightarrow$ AUD
Cross-country / cross-currency rate spillover	Ukraine/Dollar $\rightarrow$ Euro Bund, EURIBOR (Fed $\neq$ ECB)
Functional-currency / reporting-currency boilerplate	Carnival 10-Q functional currency “euro” $\rightarrow$ Euro tagged
Minority index/sector constituents	Small-cap company in a broad index proposed under case 5
Case 5 on equity indices (now sectors only)	Boeing $\rightarrow$ Nasdaq 100 (LLM hallucinated composition)
Multi-step reasoning chains	Oil sanctions $\rightarrow$ ruble $\rightarrow$ emerging-market currencies
Supply-chain-adjacent tiny exposure	China property $\rightarrow$ LME Aluminium

The exclusion list is a living document

Every rule on the previous slide started as a false positive the pipeline proposed and a human reviewer rejected. The process:

1. Run the pipeline on a new dataset.
2. Inspect accepted and rejected tags with domain experts.
3. When a pattern of incorrect proposals emerges, formalize it as an exclusion rule in the prompt.
4. Re-run and verify the rule catches false positives without killing true positives.

Open question: some exclusions may need asset-class-specific tuning:

- *Substitution reach* (case 4) may need a stricter “dominant demand driver” threshold for base metals than energy.
- *Cross-country rate spillover* may be too strict for tightly-coupled currency pairs (e.g., EUR/CHF).

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Why split into mapper and validator

(Placeholder — from ARCHITECTURE.md. Mapper recall-first, validator precision-first, base-rate shift between stages.)

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Cross-mapper agreement as a free reliability signal

(Placeholder — the 70% accept rate on `source=both` vs 13% on single-source.)

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Results on DJNW March 2022

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Aggregate numbers

(Placeholder — accepted / rejected / strong / weak / source breakdown from results/direct_djnw_v2_summary.json.)

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Focal articles walkthrough

(Placeholder — BOE Sterling, USDA Soybeans, Russia aerospace, China property, Ukraine Dollar, Nymex Crude, ICCO Cocoa, ICE Sugar.)

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Reading these slides

Each example compares the mapping set produced at the last meeting against what the current pipeline produces on the same article, under the same direct-link rules.

- **Green**: tag that the current pipeline correctly adds or keeps.
- **Orange**: tag correctly removed or still rejected.
- **Gray**: unchanged from last meeting.

The examples are chosen to show (a) the cases where iteration on the rules recovered real signal and (b) the cases where the stricter framework continues to reject chains it should reject.

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Example 1 — China property investment (Jan-Feb)

"China Jan-Feb Property Investment Rose 3.7% on Year" — slowdown in new construction starts; Hong Kong property developers mentioned.

Asset	Last meeting	Now
Hang Seng Index	–	weak 0.60 ✓
LME Copper	–	weak 0.50 ✓
Copper (High Grade)	–	weak 0.40 ✓
LME Zinc	–	weak 0.45 ✓
LME Aluminium	–	rejected
LME Lead	–	rejected
Australian Dollar	–	rejected

HSI returns via case 2 (primary listed market for China equity exposure). Copper and Zinc come in via case 4 (China construction is the dominant global demand driver for both metals).

Why copper and zinc are a real case 4, not a floodgate

Recommendation: tag LME Copper / Copper HG / LME Zinc on this article.

- The article explicitly names *“new construction starts”* and *“real estate investment in China”* — not a generic growth reference.
- Chinese construction is the *single largest* demand source for both copper (wire, plumbing) and zinc (galvanized steel). This is the textbook case-4 substitution/input link.
- The guardrail is holding: on the same article, LME Aluminium, LME Lead, and Australian Dollar are rejected as too indirect.
- Aggregate commodity FPR *fell* between versions (from 53% to 43%). If case 4 were over-firing, this would move the other way.

The rule we should hold the line on: *“dominant demand driver,” not “any industrial use.”*

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Example 2 — Dollar firms on Ukraine war

"Dollar Firms, Ukraine War Should Send It Higher" — USD strength, Fed in tightening mode, safe-haven flows into US assets.

Asset	Last meeting	Now
Euro	strong 0.85	strong 0.85
Japanese Yen	strong 0.75	strong 0.75
3-Month SOFR	weak 0.70	weak 0.70
10 Year US Government Note	weak 0.75	weak 0.75
2/5/30Y UST, Ultra T-Bond	–	weak 0.50–0.65 ✓
Euro Bund / Bobl	–	rejected
3-Month EURIBOR	–	rejected
DAX Index	–	rejected
Swedish Krona	–	rejected

The full UST curve comes in via case 3 (Fed is the issuing authority for USTs, and the article explicitly names the Fed's policy stance). Cross-country rate spillover rejections still hold cleanly.

Example 3 — BOE rate decision

“BOE’s Rate Decision Could Benefit Sterling in Short-Term”

Asset	Last meeting	Now
Sterling	strong 1.00	strong 1.00
UK Long Gilt	weak 0.60	weak 0.60
3-Month SONIA	weak 0.85	weak 0.85

Clean case 1 (Sterling named) plus case 3 (BOE is the issuing authority for gilts and SONIA). No change across versions — this is what a stable focal article looks like.

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Example 4 — The Carnival functional-currency leak

"Carnival 1Q Loss/Shr $1.66" — earnings note; the 10-Q risk-factor section lists Euro and Sterling as functional currencies.

Asset	Last meeting	Now (pre-fix)	Now (fixed)
Euro	strong 0.20	strong 0.20	rejected
Sterling	weak 0.20	weak 0.20	rejected
Consumer Discretionary	weak 0.65	weak 0.65	weak 0.65
Brent / WTI / Gas Oil	weak 0.45	weak 0.45	weak 0.45

The validator was literal-matching "euro" in the risk-factor section ("subject to foreign currency translational risk") and upgrading to strong. Case 1 already excluded "reporting currency / denominating proxy," but not prominently enough; added an explicit rejection rule to both mapper and validator prompts.

What the examples collectively show

- **Cross-jurisdictional case 2** recovered HSI on China without breaking anything else — validator chain-rejection safeguard prevents it from inventing "host country economy" intermediates.
- **Case 4** tags are well-anchored: copper/zinc on China construction pass; aluminium, lead, AUD do not. Guardrail is the word *dominant*.
- **Case 3** (issuing authority) is doing real work on rate stories — full UST curve on Fed discussion, Gilt + SONIA on BOE.
- **Stable focals** (BOE, Cocoa, Sugar, Nymex Crude, Soybeans) are unchanged, which is the expected behavior for well-specified case-1 stories.
- **Remaining leak**: functional-currency boilerplate upgraded to case 1. Fix is a one-line prompt addition to both mapper and validator.

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